CODE FILE — app.py

The actual uploadable Python source is app.py in the ZIP package. This PDF is a readable reference copy.

```
"""
Plant Leaf Health Analysis — educational prototype
Run with: streamlit run app.py
This is
not a validated disease diagnosis model.
"""
from PIL import Image
import numpy as np
import
streamlit as st

st.set_page_config(page_title="Plant Leaf Health Analysis", page_icon="■")
st.title("Plant Leaf Health Analysis")
st.write("Upload a plant-leaf image for a simple visual
color summary.")
st.warning("Educational demonstration only. This tool does not diagnose plant
disease or recommend treatment.")

uploaded = st.file_uploader("Upload a leaf photo",
type=["jpg", "jpeg", "png"])
if uploaded:
    image = Image.open(uploaded).convert("RGB")
st.image(image, caption="Uploaded leaf image", use_container_width=True)
    arr =
np.asarray(image).astype(float)
    r, g, b = arr[:, :, 0], arr[:, :, 1], arr[:, :, 2]
green_mask = (g > r * 1.08) & (g > b * 1.05) & (g > 45)
    brown_mask = (r > g * 1.12) & (g >
b * 1.05) & (r > 50)
    green_pct = 100 * green_mask.mean()
    brown_pct = 100 *
brown_mask.mean()
    st.metric("Green-toned pixels (whole image)", f"{green_pct:.1f}%")
st.metric("Brown-toned pixels (whole image)", f"{brown_pct:.1f}%")
    st.caption("These are
rough image-color measurements, not percentages of leaf area and not disease probabilities.")
    st.info("For reliable diagnosis, consult an agricultural expert and use a validated model
with a suitable labelled dataset.")
```